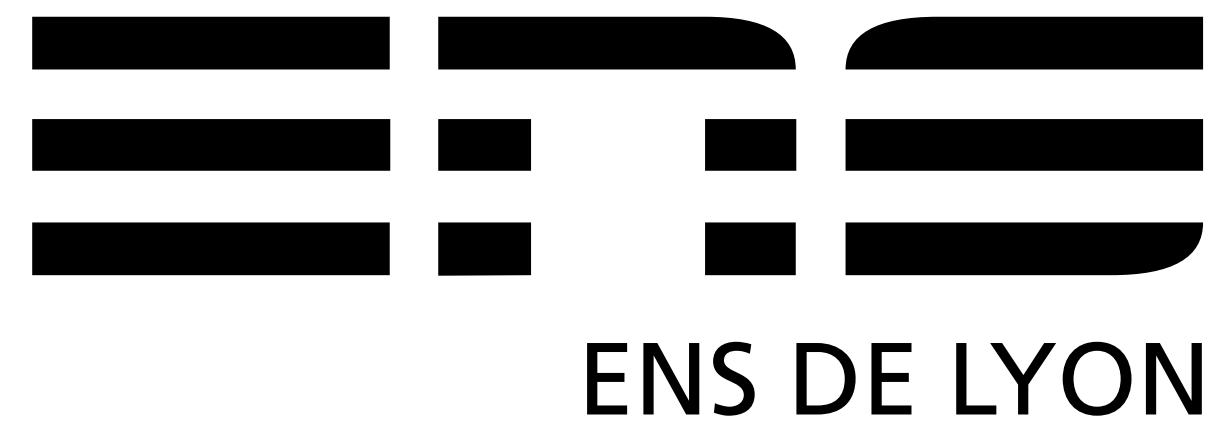


Plasma potential control in a weakly magnetized plasma column using negatively-biased emissive electrode



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OBJECTIVE

Plasma potential has been proven to be modified by a negatively-biased emissive electrode.

An extensive experimental investigation of the role of the electrode is conducted and confronted to an analytical model herein.

CONCLUSION

A dramatic effect of the electrode on the plasma potential is highlighted and relevant physical arguments are discussed.

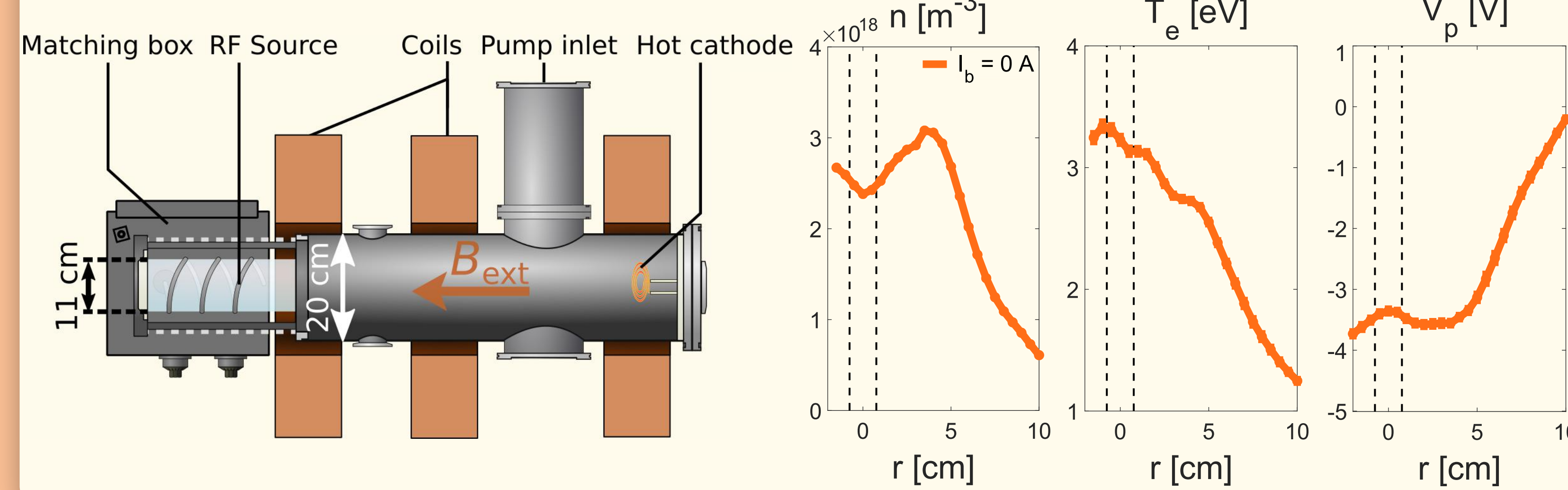
Disagreement between model and experiments may be caused by an anodic sheath unaccounted for.

1. Experimental setup

Plasma setup: RF source (13.56 MHz), magnetized [1]. A cathode faces the plasma at the end.

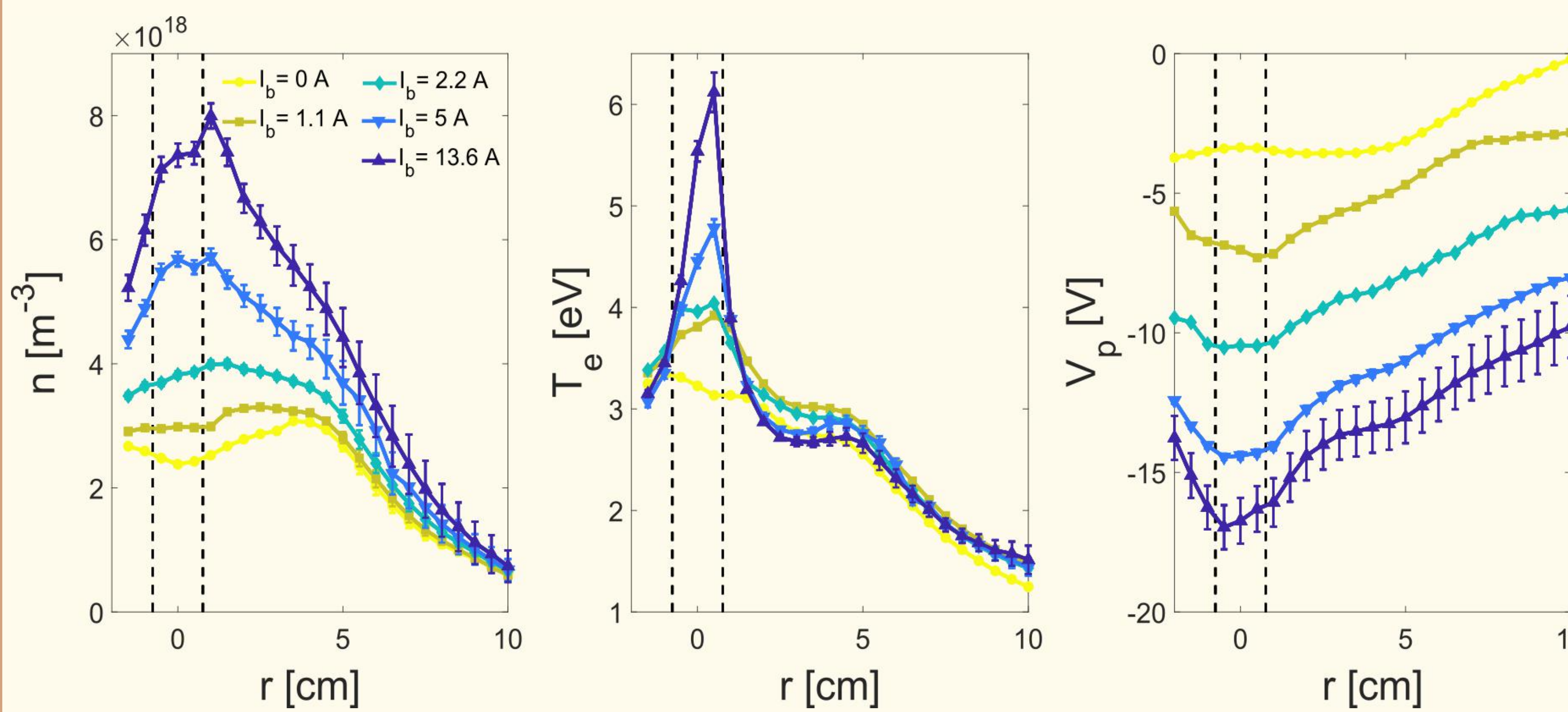
Quantity	p_0	B_{ext}	P_w	n	T_e	V_f	V_p
Value	1 – 3 mTorr	170 – 340 G	1 kW	$2 \cdot 10^{18} \text{ m}^{-3}$	3 – 4 eV	-15 V	-3 V

Diagnostics:
► Emissive probe (V_p)
► Advanced triple probe (V_f , n and T_e)
► Langmuir probe



3. Control of the plasma potential radial profile

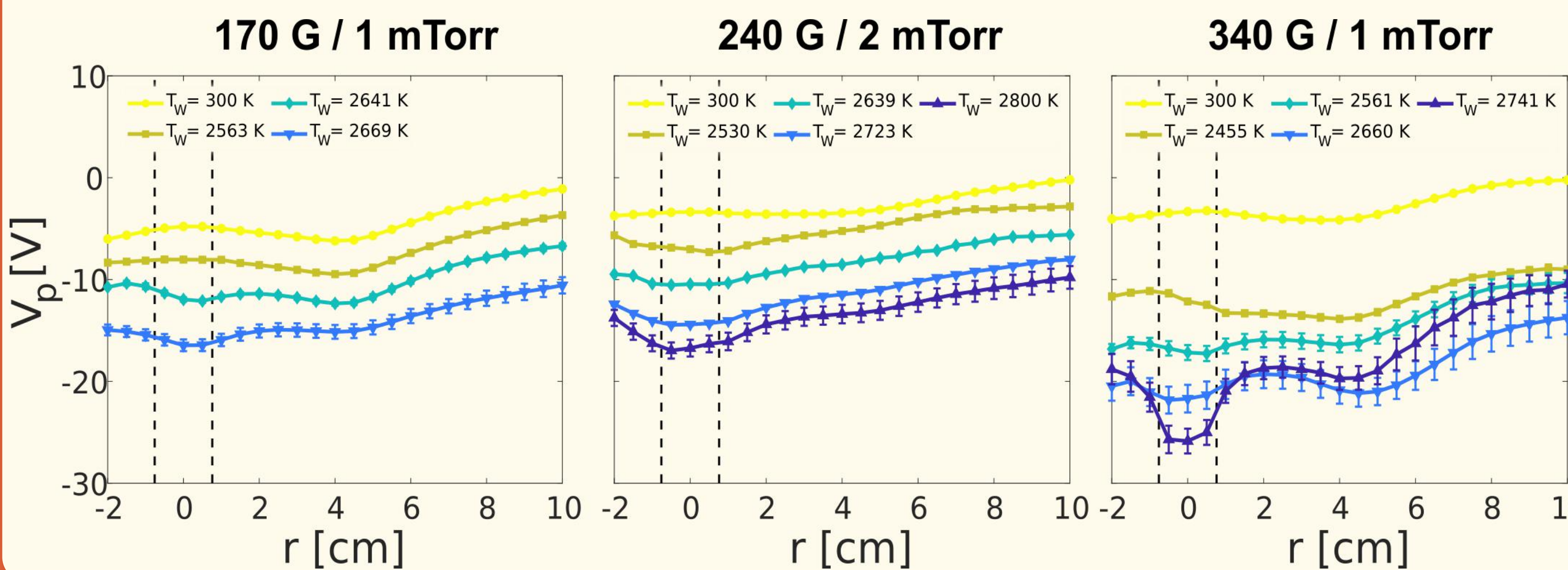
Parameters: 240 G, 2 mTorr and 1 kW. $V_b = -60 \text{ V}$ and $T_w \nearrow \Rightarrow I_b \nearrow$



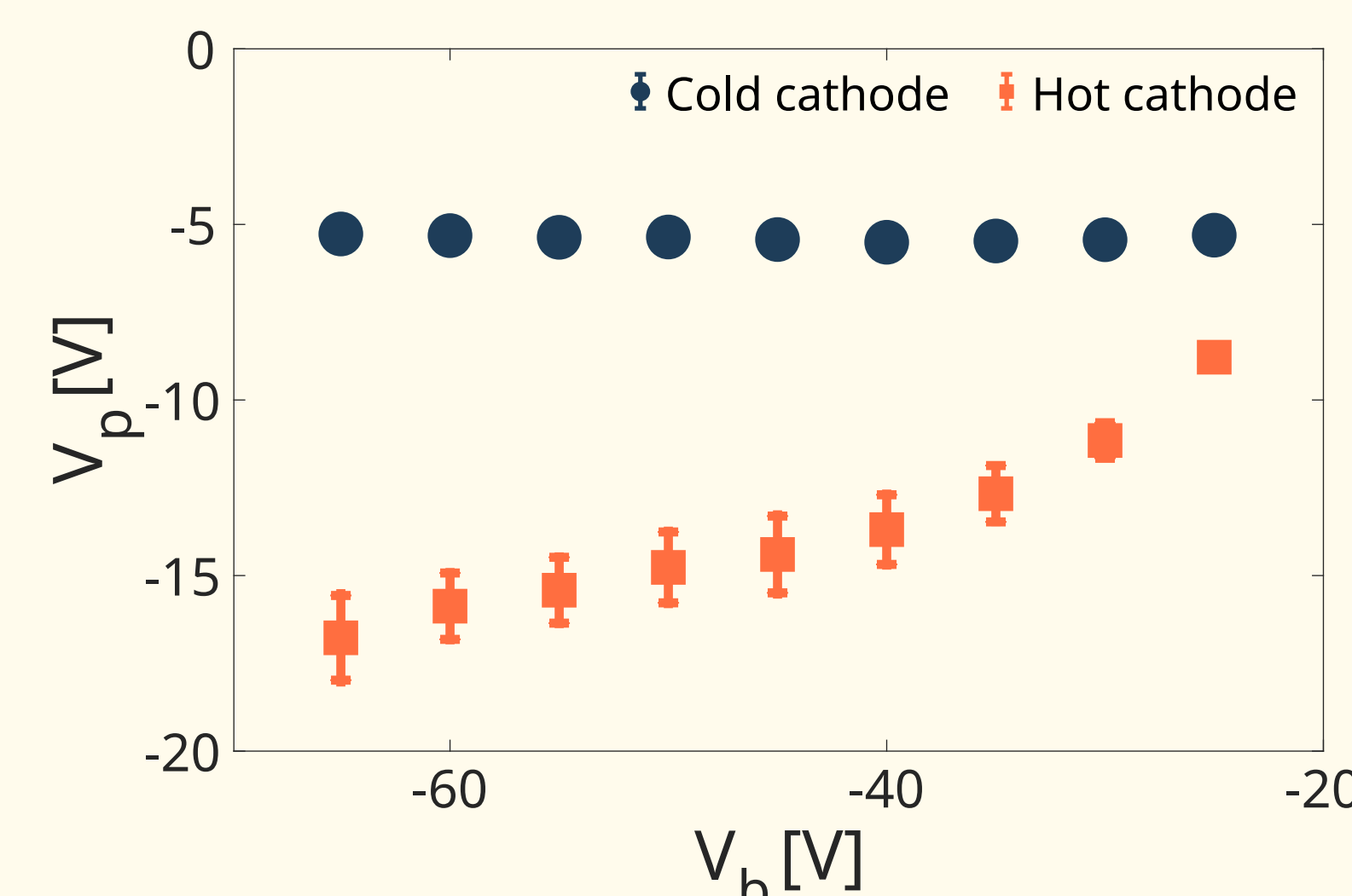
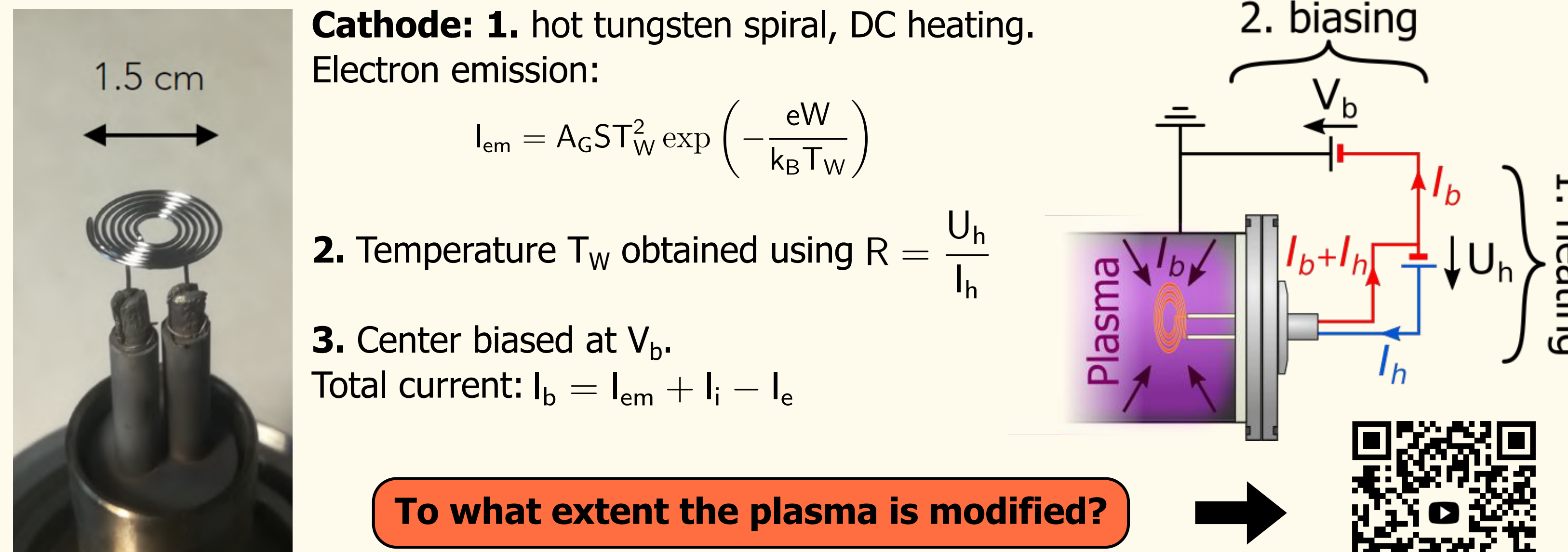
4. Influence of magnetic field and pressure

Spanning the experimental conditions:

- Enhanced influence of the cathode at high B_{ext}
- Similar behaviors between 170 G / 1 mTorr and 240 G / 2 mTorr
- V_p profiles do not evolve at high emission



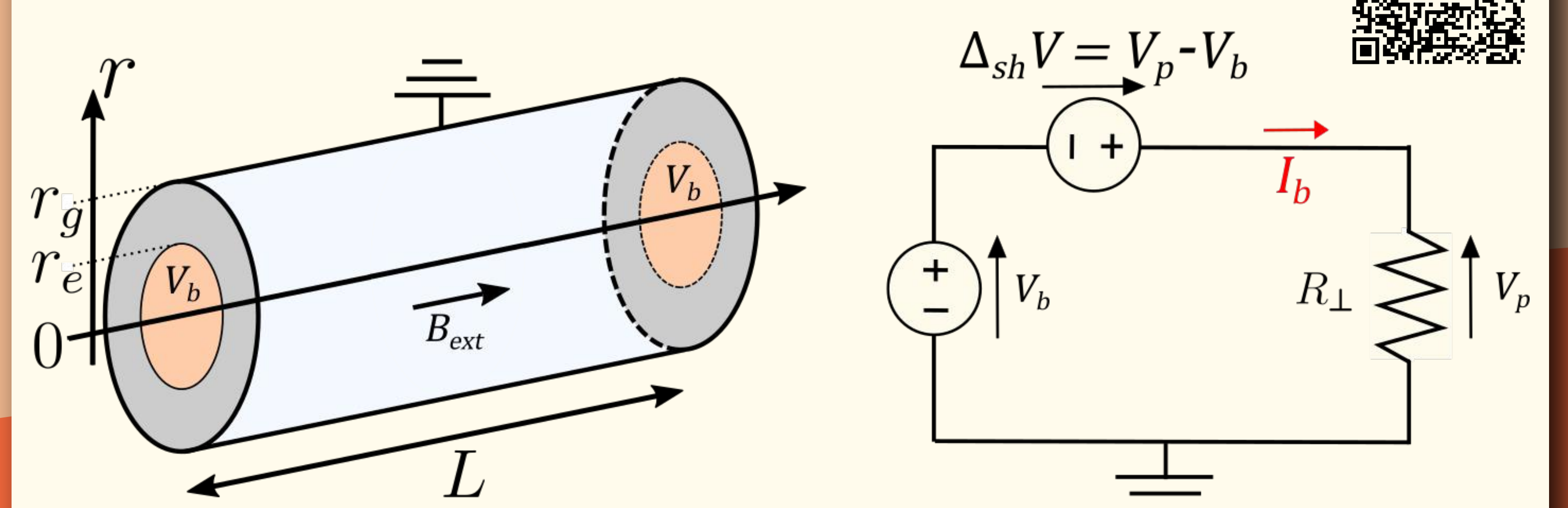
2. Effect of a negatively-biased emissive cathode



Experimental investigation:
► $V_p \searrow$ at the center when $I_b \nearrow$
► Denser and hotter plasma

5. Modeling the phenomena at stake

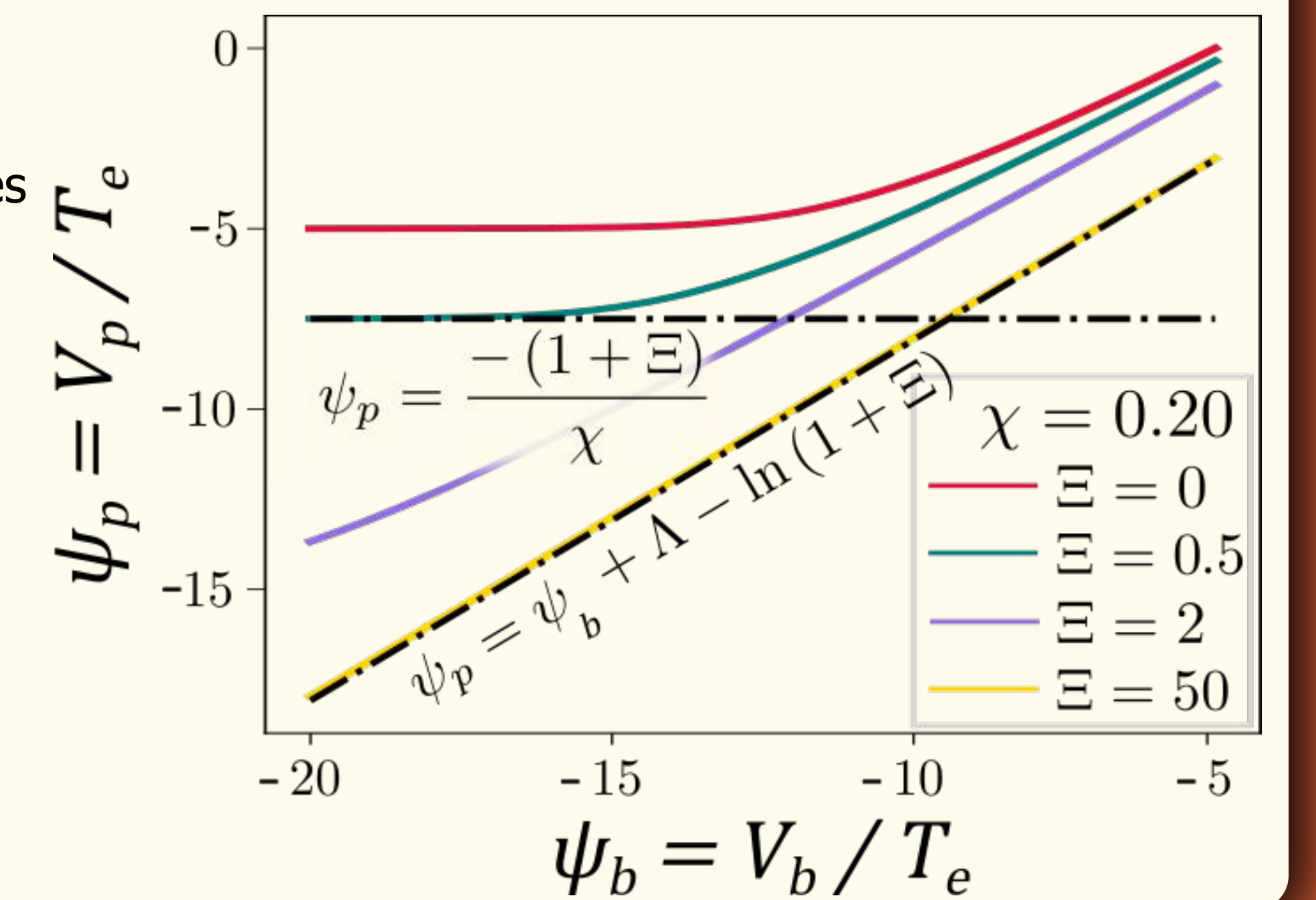
Model: Detailed in [2] Trotabas and Gueroult, Plasma Sources Sci. Technol., 2022.



Physics elements: biased cathode, cathodic sheath, anisotropic conductivities

Experimental agreement:

- Significant decrease in V_p
- Saturation regime
- $B_{ext} \nearrow \Rightarrow V_p \searrow$ for similar I_b



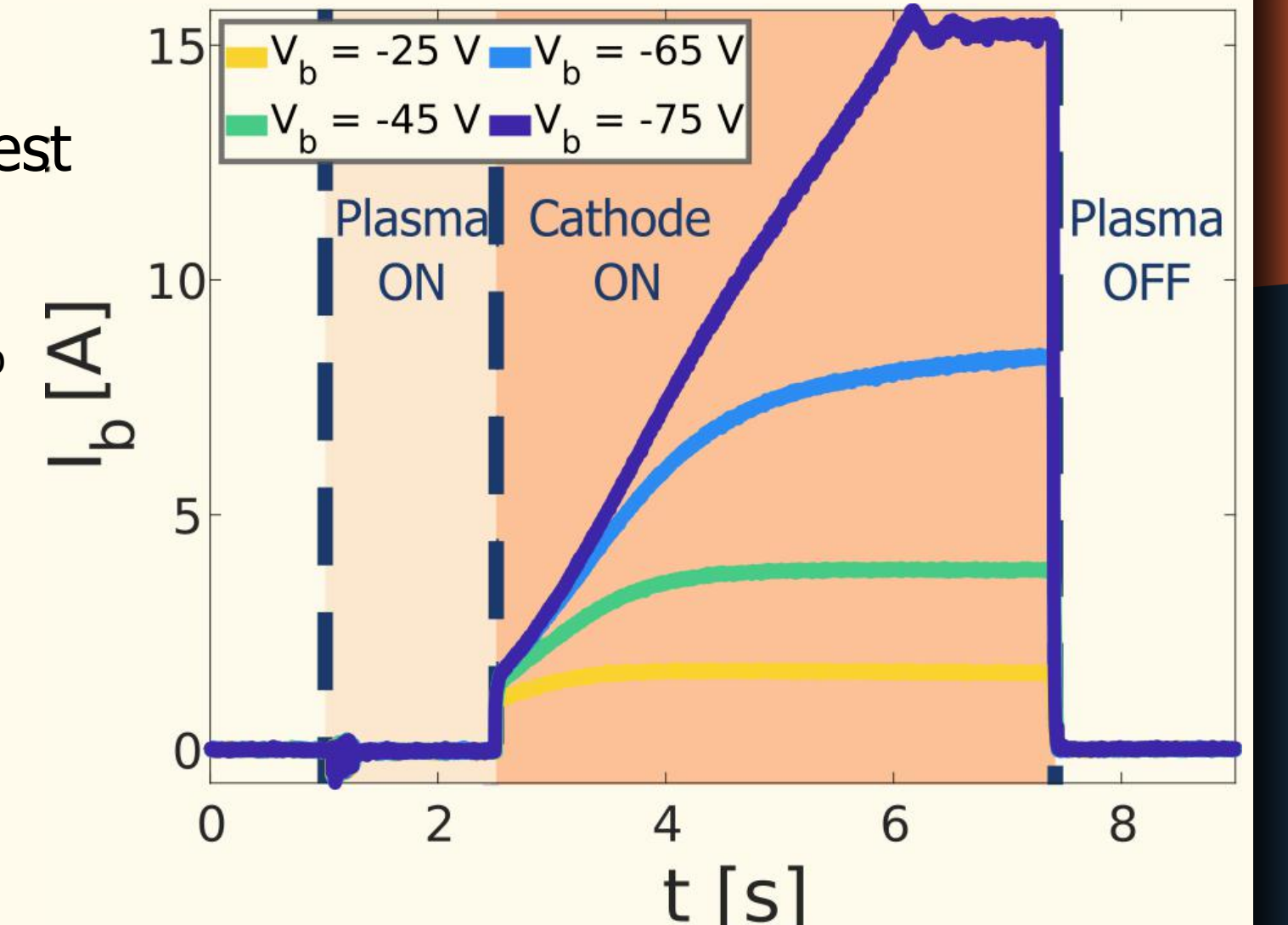
6. Limitations and prospects

Limitations:

- Only 1 cathode, plasma inhomogeneous and not at rest
- V_p near a wall $\neq 0$
- Poor agreement between measured and calculated I_b
- Unstable regime at high V_b

Prospects:

- Resolve inhomogeneous plasma numerically
- Consider an anodic sheath
- Explore the limits of the operational regime



Bibliography



[1] N. Plihon et al., J. Plasma Physics **81**, 345810102 (2015)

[2] B. Trotabas and R. Gueroult, Plasma Sources Sci. Technol. **31**, 025001 (2022)



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